FORMULA SHEET · 03

PML-AI Formula Sheet

PCI AI Project Management Leader — the complete quantitative reference

PERT · CPM · EMV · E[wait] · WSJF · PTA

PCI AI · PROJECT CONTROLS INSTITUTE GLOBAL
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AI proposes. The professional disposes.

How to use this sheet

Every formula the PML-AI Body of Knowledge teaches, in one place, with the domain where it is worked in full.

Three columns matter. **Formula** is the expression. **Meaning** is what it answers. **Watch for** is the mistake that formula attracts in practice.

This sheet is a study aid. The examination specification, including whether any formula reference is provided, is confirmed by the Institute for each published examination form.

Conventions. Durations in working periods unless stated. Ratios and indices to 2 dp. Currency in USD unless stated. Where a formula is shared with the project-controls discipline, the PCL-AI sheet is the deeper treatment and is cross-referenced.

> **The delivery-approach rule.** Several formulas below behave differently under predictive and adaptive delivery. Velocity forecasting and earned value answer the same question — when will this finish and at what cost — through different instruments. Under hybrid delivery, state which instrument governs the commitment before reporting either.

1. Notation

SYMBOL	MEANING	UNIT
`PV`, `EV`, `AC`	Planned value, earned value, actual cost	**currency**
`BAC`, `EAC`, `ETC`, `VAC`	Budget, estimate at completion, to complete, variance at completion	**currency**
`CPI`, `SPI`	Cost and schedule performance indices	**ratio**
`O`, `M`, `P`	Optimistic, most likely, pessimistic	**periods**
`tE`, `σ`	PERT expected duration, standard deviation	**periods**
`ES`, `EF`, `LS`, `LF`	Early start, early finish, late start, late finish	**date**
`TF`, `FF`	Total float, free float	**periods**
`n`	Number of stakeholders, resources or periods	**count**
`WIP`	Work in progress	**items**
`EMV`	Expected monetary value	**currency**
`PTA`	Point of total assumption	**currency**
`r`, `t`	Discount rate per period, period index	**ratio, count**

2. Business case and benefits · Domains 2, 16

FORMULA	MEANING	WATCH FOR	DOMAIN
`NPV = Σ CF_t ÷ (1 + r)^t`	Net present value of the investment	Benefits counted gross of the cost to realise them	2.3
`ROI = (Benefit − Cost) ÷ Cost`	Return on investment	Period over which benefit is counted left unstated	2.3
`BCR = PV(benefits) ÷ PV(costs)`	Benefit-cost ratio	Discounting one side and not the other	2.3
`Payback = Years before recovery + (Unrecovered ÷ Cash flow that year)`	Simple payback	—	2.3
`IRR: NPV = 0`	Internal rate of return	Multiple roots where signs change repeatedly	2.3
`Benefit realised = Actual measure − Baseline measure`	Benefits tracking	No baseline captured before delivery	16.3
`Benefits shortfall = Forecast benefit − Realised benefit`	Realisation gap	Measured too early to be meaningful	16.3

The baseline rule. A benefit cannot be measured without a pre-delivery baseline. If the baseline was never captured, the realisation number is an assertion. Capture it before delivery starts, not at handover.

3. Estimating and scheduling · Domain 6

FORMULA	MEANING	WATCH FOR	DOMAIN
`tE = (O + 4M + P) ÷ 6`	PERT expected duration (beta)	Three points from a single estimator	6.2
`tE = (O + M + P) ÷ 3`	Triangular expected duration	Confused with the PERT weighting	6.2
`σ = (P − O) ÷ 6`	PERT standard deviation	—	6.2
`σ^2 path = Σ σ^2 along the path`	Path variance	Deviations summed instead of variances	6.4
`EF = ES + Duration`	Forward pass	Off-by-one at period boundaries	6.3
`LS = LF − Duration`	Backward pass	—	6.3
`TF = LS − ES = LF − EF`	Total float	If the two forms disagree, the pass is wrong	6.3
`FF = min(successor ES) − EF`	Free float	Lags on the successor link ignored	6.3
`Critical path: TF = 0`	Longest path through the network	Constraints creating artificial zero float	6.3
`Crash cost slope = (Crash cost − Normal cost) ÷ (Normal − Crash duration)`	Cost per period saved	Crashing off the critical path	6.5
`Analogous estimate = Past × (this driver ÷ past driver)`	Top-down estimating	—	6.2
`Parametric estimate = Parameter × Rate`	Rate-based estimating	Rate from a different scope basis	6.2

Compression, compared

METHOD	MECHANISM	COST	RISK ADDED
Crashing	Add resource to critical activities	Direct cost rises	**Diminishing returns, coordination load**
Fast-tracking	Overlap activities previously in sequence	Little direct cost	**Rework risk rises materially**

4. Cost and earned value · Domain 7

Shared with the project-controls discipline. The PCL-AI Formula Sheet §4 is the deeper treatment, including earned schedule.

FORMULA	MEANING	WATCH FOR	DOMAIN
`CV = EV − AC`	Cost variance	Negative is adverse	**7.3**
`SV = EV − PV`	Schedule variance in currency	Converges to zero at completion however late	**7.3**
`CPI = EV ÷ AC`	Cost efficiency	`AC` excluding accruals	**7.3**
`SPI = EV ÷ PV`	Schedule efficiency in cost terms	—	**7.3**
`EAC = AC + (BAC − EV)`	Variance was a one-off	—	**7.4**
`EAC = BAC ÷ CPI`	Performance to date persists	—	**7.4**
`EAC = AC + (BAC − EV) ÷ (CPI × SPI)`	Cost and schedule pressure both persist	—	**7.4**
`VAC = BAC − EAC`	Variance at completion	—	**7.4**
`TCPI = (BAC − EV) ÷ (BAC − AC)`	Efficiency required to hit budget	Quoted without comparing to CPI achieved	**7.4**
`Cost to date = Actuals + Accruals`	True cost to date	The commonest omission in the discipline	**7.2**
`Resource utilisation = Productive hours ÷ Available hours`	Utilisation	Target set at 100%, which leaves no capacity for variability	**7.5**

> **Which EAC.** The formula is not the difficulty; the assumption is. Was the variance a closed one-off, does performance persist, or are cost and schedule compounding? State the assumption and publish a range rather than a point.

5. Risk and uncertainty · Domain 8

FORMULA	MEANING	WATCH FOR	DOMAIN
`EMV = Probability × Impact`	Expected monetary value	Applied to a single binary event	8.2
`Total EMV = Σ (P_i × I_i)`	Portfolio expected value	—	8.2
`Risk score = Probability rating × Impact rating`	Qualitative ranking	Ordinal ratings multiplied as if they were cardinal	8.2
`Contingency ≈ Σ EMV, or P80 − P50 from simulation`	Risk-based contingency	Percentage of budget used instead	8.3
`EV of a decision branch = Σ (P × outcome)`	Decision tree	Ignoring the cost of the option itself	8.4
`Value of perfect information = EV(with information) − EV(without)`	Whether to buy analysis	—	8.4
`Exposure after response = Residual probability × Residual impact`	Residual risk	Response cost not netted off	8.3
`P50 / P80`	Confidence levels from simulation	P80 quoted as if it were a maximum	8.3

Ordinal ratings do not multiply. A 3 × 4 = 12 heat-map score is a sorting device, not a quantity. It cannot be summed across risks, and it cannot size a contingency. Use expected value or simulation for anything that touches money.

6. Agile, adaptive and flow · Domain 13

FORMULA	MEANING	WATCH FOR	DOMAIN
`Velocity = Points completed ÷ Sprint`	Delivery rate	Best-ever used rather than rolling average	13.3
`Sprints remaining = Points remaining ÷ Average velocity`	Forecast to completion	Backlog growth ignored	13.3
`Capacity = Team members × Available days × Focus factor`	Sprint capacity	Focus factor set at 1.0	13.3
`Cycle time = WIP ÷ Throughput`	Little's Law	Only valid for a stable system over the window measured	13.4
`Throughput = Items completed ÷ Period`	Delivery rate in items	—	13.4
`Lead time = Cycle time + Queue time`	Customer-observed duration	Reported as cycle time	13.4
`Flow efficiency = Active time ÷ Total elapsed time`	Proportion of time actually worked	—	13.4
`WSJF = Cost of delay ÷ Job size`	Weighted shortest job first	Cost of delay guessed rather than derived	13.3
`% complete = Points completed ÷ Total planned points`	AgileEVM progress	Point inflation across sprints	13.5
`Run rate = Team cost ÷ Sprint`	Cost per sprint	—	13.5
`Burn-up: scope line and completed line plotted together`	Progress against a moving scope	Burn-down hides scope growth; burn-up does not	13.3

Why burn-up beats burn-down for governance. A burn-down chart shows work remaining, so scope added mid-flight looks like slow progress. A burn-up plots completed work and total scope as two lines, which makes scope growth visible as what it is. For any project reporting to a gate, use burn-up.

7. Procurement and contracts · Domain 10

FORMULA	MEANING	WATCH FOR	DOMAIN
`PTA = [(Ceiling price − Target price) ÷ Buyer share ratio] + Target cost`	Point of total assumption (FPIF)	Seller's share ratio used in the denominator	10.3
`Fee = Target fee + Share ratio × (Target cost − Actual cost)`	Incentive fee	Sign convention on an overrun	10.3
`Final price = Actual cost + Fee` (capped at ceiling)	FPIF settlement	Ceiling not applied	10.3
`Pain/gain = Share ratio × (Actual − Target)`	Target-cost mechanism	—	10.3
`LD exposure = LD rate × Days late`	Liquidated damages	Cap not applied	10.4
`Make-or-buy: compare total cost of ownership, not price`	Sourcing decision	Transition and exit costs excluded	10.2
`Weighted score = Σ (criterion weight × score)`	Tender evaluation	Weights set after seeing the bids	10.2

What the PTA means. Above the point of total assumption, the seller absorbs every further dollar of cost — the buyer's contribution has reached the ceiling. It is the point at which the contract's incentive stops working and the seller's interest changes character. **Check:** at an actual cost exactly equal to the PTA, the computed final price equals the ceiling price precisely. If it does not, the share ratio has been applied the wrong way round.

8. Quality and assurance · Domain 9

FORMULA	MEANING	WATCH FOR	DOMAIN
`Cost of quality = Prevention + Appraisal + Internal failure + External failure`	Total quality cost	Only failure costs counted	9.2
`DPMO = (Defects ÷ (Units × Opportunities)) × 1,000,000`	Defects per million opportunities	Opportunity count defined inconsistently	9.3
`Defect density = Defects ÷ Size`	Quality rate	—	9.3
`Rework ratio = Rework effort ÷ Total effort`	Rework burden	—	9.3
`Control limits = Mean ± 3σ`	Statistical process control	Control limits confused with specification limits	9.3
`Process capability = (USL − LSL) ÷ 6σ`	Capability index	—	9.3
`First-pass yield = Units passing first time ÷ Units started`	Right-first-time	—	9.3

Control limits are not specification limits. Control limits describe what the process does; specification limits describe what the customer requires. A process can be in control and incapable at the same time, and the response to each is different.

9. Stakeholders, teams and communication · Domains 11, 12

FORMULA	MEANING	WATCH FOR	DOMAIN
`Channels = n(n − 1) ÷ 2`	Communication channels among `n` people	Growth is quadratic, and this is the point	11.2
`Δ Channels = n` when one person joins a team of `n`	Marginal communication load	—	11.2
`Influence/interest grid position`	Stakeholder strategy	Assessed once and never revisited	11.1
`RACI: exactly one A per activity`	Accountability rule	More than one A means none	12.2
`Team productivity ≠ Σ individual productivity`	Coordination overhead	—	12.3

Worked check. A team of 9 has 9 × 8 ÷ 2 = **36** channels. Adding one person takes it to 45 — nine new channels for one new person. This is the arithmetic behind small-team preference, and it is why adding people to a late project rarely accelerates it.

10. Programmes and portfolios · Domain 15

FORMULA	MEANING	WATCH FOR	DOMAIN
`Portfolio value = Σ NPV of component projects`	Aggregate value	Interdependencies ignored	15.2
`Scoring model = Σ (criterion weight × criterion score)`	Prioritisation	Weights adjusted to justify a chosen answer	15.2
`Capacity constraint: Σ resource demand ≤ Available capacity`	Portfolio feasibility	Demand summed at 100% utilisation	15.3
`Programme EV = Σ component EV`	Aggregated performance	Components on different measurement bases	15.4
`Benefit dependency: benefit realised only when all enablers delivered`	Benefits logic	Partial delivery credited with full benefit	15.4

11. Worked micro-examples

Each is internally consistent and reproducible from this sheet.

PERT. `O` = 8, `M` = 12, `P` = 22.

- `tE` = (8 + 48 + 22) ÷ 6 = **13.0**
- `σ` = (22 − 8) ÷ 6 = **2.33** · `σ^2` = **5.44**
- Triangular for comparison = (8 + 12 + 22) ÷ 3 = **14.0** — a different method, a different answer

Float. `ES` = 4, duration 6, `LF` = 14.

- `EF` = 10 · `LS` = 8 · `TF` = 8 − 4 = **4** = 14 − 10 — the two forms agree

Point of total assumption. Target cost 100,000 · target price 113,000 (fee 13,000) · ceiling 120,000 · buyer share 80%.

- `PTA` = [(120,000 − 113,000) ÷ 0.80] + 100,000 = 8,750 + 100,000 = **108,750**
- Check at `AC` = 108,750: overrun 8,750; seller absorbs 20% = 1,750; fee = 13,000 − 1,750 = 11,250; price = 108,750 + 11,250 = **120,000**, exactly the ceiling price

Earned value. `PV` 168,000 · `EV` 151,200 · `AC` 170,400 · `BAC` 420,000.

- `CPI` = **0.887** · `SPI` = **0.900**
- `EAC` range = **439,200** (one-off) to **473,300** (CPI persists) to **507,000** (both persist)
- `TCPI` to BAC = (420,000 − 151,200) ÷ (420,000 − 170,400) = **1.08** against a CPI of 0.89 — the budget requires a 22% efficiency improvement on work that has so far run 11% below plan

Little's Law. `WIP` = 12 items, throughput = 4 items per week.

- Cycle time = 12 ÷ 4 = **3 weeks**. Halving WIP to 6 halves cycle time to 1.5 weeks at the same throughput.

Communication channels. `n` = 9 → **36** channels. `n` = 10 → **45**.

12. The ten formulas most often misapplied

1. **Risk scores multiplied and summed.** Ordinal heat-map ratings are a sorting device. They cannot size a contingency.
2. `SV` **used as a schedule measure late in a project.** Denominated in currency; converges to zero at completion however late. Use earned schedule.
3. **PTA computed with the seller's share ratio.** The denominator is the **buyer's** share. Check that the price at PTA equals the ceiling.
4. **Velocity taken as best-ever.** Use a rolling average across recent, comparable sprints.
5. **Burn-down used for gate reporting.** It conceals scope growth. Burn-up shows both lines.
6. **Standard deviations summed along a path.** Variances add; take the square root at the end.
7. **Crashing an activity off the critical path.** Cost incurred, no duration saved.
8. **Control limits confused with specification limits.** In control and incapable are different states with different responses.
9. **Resource utilisation targeted at 100%.** Leaves no capacity to absorb variability; queues grow non-linearly as utilisation approaches full.
10. **Benefits claimed without a pre-delivery baseline.** Without the baseline the realisation figure cannot be computed, only asserted.

13. Cross-references

FOR THE FULL TREATMENT	SEE
Every formula worked with figures and exercises	**The PML-AI Body of Knowledge, at the domain cited**
Earned value, earned schedule and forecasting in depth	**PCL-AI Formula Sheet, §4**
Investment appraisal and cost of capital in depth	**PFL-AI Formula Sheet, §3 and §4**
Why forecasts fail to change decisions	**Knowing Early, Knowledge Series 01, §5**

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